

Summary

This paper studies **failure attribution for LLM-based agents**, specifically how to identify the *decisive error step* from failed trajectories without using step-level labels for constructing the reference set. The proposed method, **SOAP (Spectral scOring with Attention-guided Propagation)**, has two components: (1) a spectral base scorer that builds an SVD-based reference space from representations of unlabeled failed trajectories and assigns larger anomaly scores to steps with low projection energy in a selected spectral band; and (2) an attention-guided rescoring mechanism that propagates strong downstream failure evidence back toward earlier steps on which those downstream steps attend, addressing what the paper calls **downstream contamination**. [turn0file0 · L62-L70] [turn0file0 · L192-L205]

The method is evaluated on WHO&WHEN, CORRECT-ERROR, and TRACEELEPHANT, covering five subsets, using Qwen3.5-9B and DeepSeek-R1-Distill-Llama-8B as primary backbones. SOAP substantially outperforms prompt-based and training-based baselines in the main experiments; for example, with Qwen3.5-9B it obtains 47.62/34.48/61.78/35.66/23.19 step accuracy across the five subsets, while the corresponding DeepSeek results are 45.50/28.74/64.68/42.64/30.43.

[turn0file0 · L722-L757] The paper further includes larger-backbone experiments, cross-distribution transfer, synthetic reference trajectories, scoring-function comparisons, rescoring ablations, sensitivity analyses, top-*k* evaluation, and runtime analysis.

Strengths

- 1. The paper addresses a practically meaningful and well-motivated setting.** Using failed trajectories without requiring step-level annotations for constructing the reference set is attractive because task failure is often automatically observable, whereas manually locating the decisive error is expensive. The distinction between an initial error and later symptoms is also important for long-horizon agent debugging. [turn0file0 · L87-L91] [turn0file0 · L177-L191]
- 2. The empirical results are consistently strong and unusually comprehensive.** SOAP improves over several families of baselines across five subsets and two primary backbones, while the additional experiments examine larger models, synthetic references, cross-distribution transfer, top-*k* retrieval, and different amounts of reference data. The ablations are particularly useful: the spectral band strongly outperforms perplexity, norm-based, top/tail/full-spectrum alternatives, and attention-guided rescoring improves over uniform and position-based alternatives. [turn0file0 · L1044-L1062] [turn0file0 · L1075-L1100]
- 3. The method is relatively simple and computationally attractive compared with training-heavy or multi-call approaches.** SOAP does not fine-tune the backbone, and the paper provides extensive implementation details, sensitivity analysis, seed variance, and runtime comparisons. The reported SVD fitting cost is very small relative to OAT and StepFinder, while inference avoids expensive multi-stage LLM judging. [turn0file0 · L2021-L2048]

Drawbacks

1. The most important concern is that the paper's **“without step-level labels” characterization is somewhat stronger than what the experimental protocol actually supports**. The reference split indeed hides step annotations, but the data are split 30/20/50 into reference/validation/test, and the authors explicitly state that **validation step labels are used for hyperparameter selection**. Moreover, SOAP selects the representation layer, spectral band, attention band, and propagation strength using this labeled validation set. [turn1file0 · L756-L766] This matters because the spectral search itself considers **210 candidate contiguous bands**, in addition to the layer, predecessor window, attention band, and γ . [turn1file0 · L1712-L1729] Thus, SOAP is more accurately described as *step-label-free for fitting the reference representation*, rather than fully step-label-free. I would like to see the main results reproduced with a single globally fixed configuration, or with an entirely label-free model-selection criterion. This experiment would materially strengthen the paper's central claim.
2. Relatedly, the **cross-distribution experiment is not a completely label-free or zero-shot transfer experiment**. Although the main text describes constructing the reference on one source subset and evaluating on another, the appendix clarifies that for every source-target pair the configuration is **re-selected on the target validation split**, and in the cross setting the target reference and validation files are pooled for this purpose. [turn1file0 · L2881-L2894] Consequently, Figure 4(c)/Figure 9 demonstrates transfer of the *spectral reference*, but not transfer of the complete SOAP configuration. A substantially stronger generalization experiment would fit both the reference and hyperparameters exclusively on the source domain and transfer everything unchanged to the target.
3. The proposed mechanism is empirically effective, but its interpretation remains somewhat heuristic. In particular, the argument that decisive-error information should reside in a particular intermediate spectral band is not theoretically established, and the favorable band is selected using labeled validation data. Similarly, attention mass is treated as a proxy for step dependency; the rescoring ablation shows that attention works better than uniform or positional alternatives, but it does not directly demonstrate that the extracted attention graph corresponds to genuine causal/information dependencies. This distinction is relevant because dependency-aware root-cause tracing itself is not entirely new: GraphTracer explicitly distinguishes downstream symptoms from upstream causes through dependency graphs [1], while MASPrism uses NLL and attention signals to move from symptom-like steps toward earlier candidate sources [2]. SOAP's clearest novelty therefore appears to be the **spectral reference constructed from unlabeled failure trajectories and its integration with lightweight attention propagation**, rather than the broader downstream-contamination insight itself. [turn0file0 · L1193-L1210] (arxiv.org)
4. The comparison involves **different data-access regimes across methods**, which makes some of the paper's conclusions stronger than the experiment strictly warrants. SOAP receives target-domain failed reference trajectories plus labeled target validation trajectories; OAT is trained on 103 successful MCP-Atlas trajectories, while StepFinder uses separately released step-labeled corpora of 1,564 CaptainAgent and 2,604 Magentic-One trajectories. [turn1file0 · L1841-L1857] Therefore, the observation that SOAP beats supervised methods is compelling as a performance result, but it does not by itself demonstrate that unlabeled failure data are intrinsically more informative than step-level supervision—the methods have

materially different data distributions, architectures, and adaptation protocols. A matched-data or matched-validation-budget comparison would make this conclusion considerably stronger.

References for the prior-work discussion

[1] *GraphTracer: Graph-Guided Failure Tracing in LLM Agents for Robust Multi-Turn Deep Search*. Heng Zhang et al., arXiv, 2025. (arxiv.org)

[2] *MASPrism: Lightweight Failure Attribution for Multi-Agent Systems Using Prefill-Stage Signals*. Yang Liu et al., arXiv, 2026. (arxiv.org)

Rating

Rating: 6/10 — Marginally above the acceptance threshold.

My assessment is closer to **6 than 5** because the core empirical contribution is substantial: the method is simple, the results are consistently strong across datasets and backbones, and the experimental section contains considerably more analysis than a typical borderline paper. In particular, the spectral-scoring and attention-rescoring components are both supported by meaningful ablations.

I would not currently give **7 or 8**, primarily because the strongest conceptual claim—failure attribution “without step-level labels”—is complicated by extensive target-specific selection using labeled decisive-error validation data. The cross-distribution experiment has the same issue. If the authors show that a **fixed or completely unsupervised hyperparameter configuration preserves most of the gains**, I would be comfortable moving the score to **7**. Conversely, if performance drops substantially without target step-label tuning, I would lean toward **5**.

Confidence: 4/5. I am fairly confident in the methodological and experimental assessment, although some very recent failure-attribution literature is contemporaneous with this submission.

Prompt-injection check: I did not detect reviewer-directed or manipulative instructions in the manuscript.